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Review

Using appropriate methods to assess population exposure to extreme weather events: Why and how?

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Abstract

The existing assessment of population exposure to extreme weather events has established various standards, particularly in the definitions of extreme weather events and the methods of population exposure assessment, hindering reliable risk perception. This study reviews three core threshold types for defining extreme weather events (absolute, relative and composite) and three major population exposure assessment methods (static, dynamic and alternative indicator-based), analysing their advantages, disadvantages and applicability. These findings demonstrate that the multi-factorial composite threshold can reflect the health-related risks of extreme weather events from a multi-dimensional perspective, promoting a paradigm shift from 'weather-centric theory' to 'impact-centric theory' in related research. Meanwhile, dynamic population exposure assessment based on real-time spatiotemporal behavioural responses of the population is the core innovation direction to break through the limitations of static assessment and improve the accuracy of exposure quantification. This study also identifies key research limitations in the current field, including non-standardised and non-multi-factorial definitions of extreme weather events, methodological biases inherent in exposure assessment, inadequate integration of population vulnerability and adaptability dimensions and limited applicability of the assessment system in data-scarce regions. Thus, by clarifying the necessity of scientific assessment and adopting targeted strategies—promoting the standardisation and multi-factorial transformation of extreme weather event definitions, optimising the methodological system of population exposure assessment and constructing an integrated assessment framework of exposure−vulnerability−adaptability—we can bridge the gap between scientific research and practical application, and improve the scientific rigour and effectiveness of the assessment of population exposure to extreme weather events.

Keywords: Extreme weather event; Population exposure; Definition; Risk assessment

1. Introduction

Global climate warming has become an undeniable reality: the global average surface temperature increased by 1.09 °C [0.95−1.20 °C] between 1850−1900 and 2011−2020 (Intergovernmental Panel on Climate Change (IPCC, 2023), and a new daily global average temperature record of 17.16 °C was set in July 2024 (ECMWF, 2024). This has intensified extreme weather events around the world, as manifested by heatwaves, droughts, out-of-season wildfires, floods and compound events (Mallapaty, 2022; WMO, 2024). Extreme weather events have become the new normal today and into the future, eliciting a global response (Ogunbode et al., 2019; Trenberth et al., 2015).

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According to the report 2024 Disasters in Numbers, the Emergency Events Database (EM-DAT) recorded 393 natural hazard-related disasters in 2024. Among them, the indicator of total deaths caused by meteorological disasters (floods, extreme temperatures and storms) exceeded 80% (13,712 fatalities) of the indicator of total deaths caused by natural hazard-related disasters (EM-DAT, 2025). In addition to their catastrophic consequences, extreme weather events indirectly affect individuals' health and well-being through physical illnesses, mental health disorders, water contamination and infectious diseases, imposing obvious economic and health burdens on society (Franchini and Mannucci, 2015; Wolf et al., 2015; Friel et al., 2011). The risk of an extreme weather event depends not only on the hazard itself but also on the degree of exposure to it, the vulnerability of individuals and communities and their capacity to prepare for, manage and recover from extreme events (Ebi et al., 2021). Health departments, emergency management authorities, urban planners and community service providers need to prepare for the disasters and adverse impacts caused by extreme weather events.

Understanding population exposure to extreme weather events is a prerequisite for mitigating the adverse effects. Over the past decades, great progress has been made in defining extreme weather events and developing population exposure assessment methods, laying a solid foundation for risk management and policy formulation, but the various definition standards for extreme weather events lead to substantial heterogeneity in assessment outcomes. Existing definitions are primarily categorised into three threshold types, each with distinct applications. Because they are straightforward and intuitive, absolute threshold indicators—based on fixed meteorological values—are widely used for preliminary exposure screening and emergency resource allocation (National Disease Control and Prevention Administration (NDCPA, 2023). Relative threshold indicators, derived from percentile critical values of local historical climate data, have become the standard for cross-regional comparisons by addressing the limitations of absolute thresholds in capturing regional climate variability (Tang and Hu, 2022). More recently, composite threshold indicators have emerged as a revolutionary advance, integrating multiple meteorological elements, spatiotemporal factors and even health data to identify extreme weather risks from a multi-dimensional perspective (Li et al., 2013).

Static population exposure assessment methods—predicated on stable population distribution assumptions—have been widely used for large-scale preliminary risk screening and macro-level public health planning (Liu et al., 2023; Grant et al., 2025). Dynamic assessment methods, integrating real-time spatiotemporal data, overcome the constraints of static methods by capturing population mobility and environmental heterogeneity (Xie et al., 2020; Zhang et al., 2024). Furthermore, alternative indicators have provided a viable alternative for regions with limited direct data, indirectly capturing both exposure and vulnerability dimensions (Russo et al., 2019; Smith and Katz, 2013; Keim, 2020).

Notwithstanding these advances, current research into population exposure to extreme weather events still faces great challenges. Among these are the inherent limitations in accuracy and applicability of the various definition standards for extreme weather events and existing population exposure assessment methods. Consequently, there is a lack of consensus among researchers about which definitions and methods to use in specific contexts. By analysing definitions of extreme weather events and the relative strengths of population exposure assessment methods, this study aims to clarify which methods for population exposure assessment in extreme weather are most effective in specific contexts, providing a roadmap for future assessments and ultimately enhancing their scientific rigour and effectiveness.

2. Definitions of extreme weather events

Defining extreme weather events is fundamental to population exposure assessment, as it directly determines the accuracy of exposure identification and the reliability of assessment outcomes. The Expert Team on Climate Change Detection and Indices (ETCCDI) has defined 27 representative climate indices, which promoted global observations and research into extreme weather events, accelerating the progress of simulation and attribution studies. However, the lack of a standard definition has led to heterogeneity in population exposure assessment methods. Existing definitions are mainly categorised into three types of threshold indicators: absolute, relative and composite. This study summarises the core characteristics, advantages and disadvantages of each type, with a specific focus on their implications for population exposure assessment (Fig. 1).

2.1. Absolute threshold indicators

Absolute threshold indicators define extreme weather events based on a fixed threshold value of meteorological elements that have obvious impacts on humans or organisms. Their core advantage lies in their simplicity and intuitiveness, which facilitate calculation and policy dissemination. Consequently, these indicators are particularly suitable for preliminary exposure estimation in climatically homogeneous regions, enabling rapid screening of the scale of exposed populations within administrative regions to support initial resource allocation during disaster emergencies. For example, in China, a yellow alert for high temperatures is triggered when the daily maximum temperature reaches or exceeds 35 °C for more than three consecutive days (NDCPA, 2023).

Nevertheless, when integrated into population exposure assessment, absolute thresholds struggle to account for regional heterogeneity. Due to differences in long-term climate conditions and population adaptability across

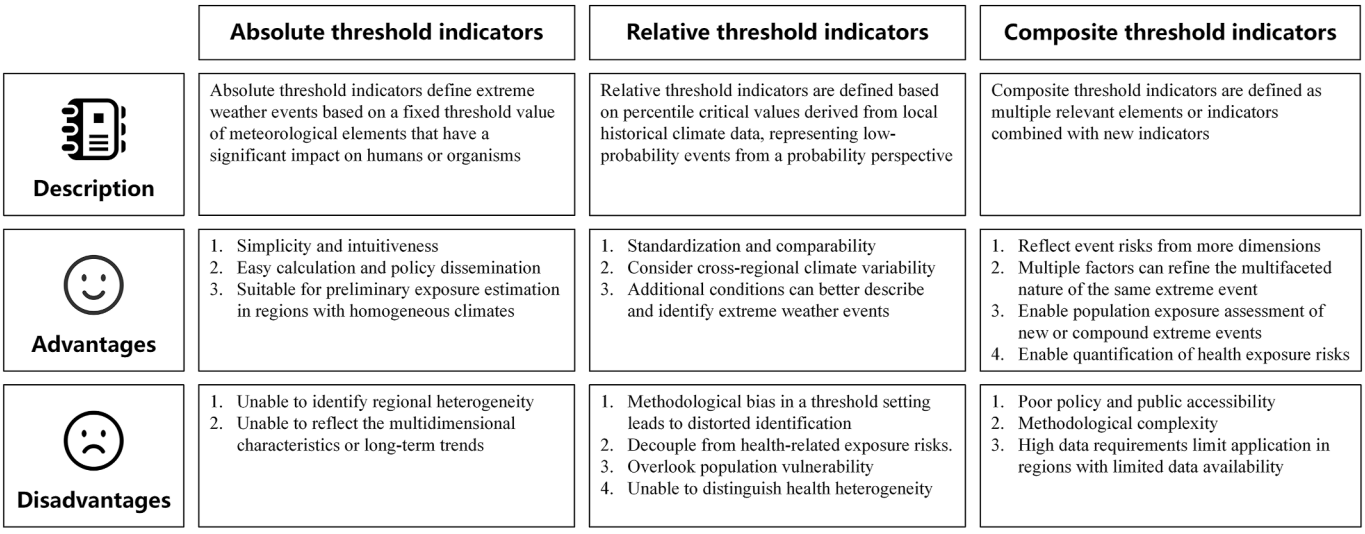


Fig. 1. Advantages and disadvantages of threshold indicators.

regions, applying a uniform absolute threshold can introduce bias into exposure assessments. Consequently, some researchers have adapted the official definitions of various extreme weather events to align with local climate characteristics. For example, the definition of a heatwave considers the duration of high-temperature weather. Based on Xinjiang's climatic characteristics—high summer temperatures, intense solar radiation and low relative humidity—high-temperature days were defined as those with a daily maximum temperature $\geq$ 35 °C in Xinjiang in 2001–2020 and classified as mild, moderate and severe grades, corresponding to continuous high-temperature periods of 3–4, 5–7 and $\geq$8 d (Dong et al., 2022). This example illustrates how modified absolute threshold indicators based on official definitions and local climate characteristics can improve local applicability and can be used in the initial exploration stage of population exposure assessment to estimate the size of a population exposed to a given extreme weather event. However, such adjustments do not address the fundamental issue of threshold incompatibility in cross-regional exposure assessments. In addition, absolute thresholds are static and cannot reflect the multi-dimensional characteristics or long-term trends of extreme weather events, leading to distortions in dynamic exposure assessments (e.g. interannual changes in exposure levels). To better address the above two limitations, using relative threshold indicators to characterise extreme weather events is currently the mainstream approach in population exposure assessment studies of extreme weather events.

2.2. Relative threshold indicators

Relative threshold indicators are defined based on percentile critical values derived from local historical climate data, representing low-probability events from a probabilistic perspective. This approach considers climate variability across regions, the absolute intensities of which avoid the issue of difficulty in comparing extreme weather events with uniform standards.

Before determining relative threshold indices, it is essential to consider observational data and reference periods. Due to the increasing maturity of current meteorological observation technology, most studies use daily data, while a few still use monthly or annual data. For example, the probabilistic distribution of daily maximum/minimum temperature data corresponds to relatively hot/cold events during the day and night, emphasising extreme moments within a day, identifying heatwaves and coldwaves effectively. However, for regions with large daily temperature variability, it overestimates population exposure time. Conversely, the daily mean temperature reflects the average temperature state each day, but underestimates the severity of exposure in regions with wide daily temperature variability. Therefore, the choice of data as a sample depends on research objectives and data availability.

Regarding reference periods, two forms are mainly used (Folland et al., 1999; Li et al., 2013): 1) A 30-year baseline climate period is specified by the WMO as the climate reference period, especially in predictive studies. When the mean values of climate indicators between two periods do not differ significantly, extreme weather event detection becomes less sensitive. 2) The entire study period is taken as the reference period, especially in retrospective or current studies. The convergence of average relative threshold values across different regions towards the mean of a typical recurrence period impedes a comparison of the trends of extreme weather events across regions. Regardless of the method chosen, the limitations of the reference period itself must be addressed. While static reference periods may systematically underestimate the frequency and intensity of current extreme events, rolling reference periods—though able to capture the trend of extreme events—make long-term series comparison meaningless.

The core advantage of the relative threshold method is that percentile thresholds can promote standardisation and comparability. When choosing relative threshold indicators for population exposure assessment, the 95th (5th) percentile of a meteorological element for a period of time in a research area or grid is often used to define corresponding extreme weather events (Tang and Hu, 2022). Additionally, researchers can enhance methodological flexibility by adjusting percentiles (e.g. 90th, 95th, 99th) and duration criteria to match specific exposure assessment scenarios and capture region-specific exposure risks (Founda et al., 2022; Sun et al., 2022; Watanabe et al., 2022). Based on these percentile thresholds, additional conditions adapted to different extreme weather events have been derived to better describe and identify extreme weather events. For example, when defining extreme precipitation, to exclude regional and seasonal factors, wet days (daily precipitation $\geq$ 1 mm) are predefined, and the 95th percentile is selected from the wet-day sequence (Liu et al., 2020). For regions where thresholds are too low or high, leading to unreasonable extreme weather events, an absolute threshold is added, for example, setting 25 °C as the extreme high-temperature threshold when the 95th percentile of daily maximum temperatures is below 25 °C (Liu et al., 2017).

However, the use of relative indicators for population exposure assessment presents distinct limitations. First, the methodological bias in threshold setting leads to the distorted identification of extreme events. The mainstream percentile threshold method frequently uses a running window to increase the sample size for the threshold calculation. However, running windows introduce a time-, region- and dataset-dependent bias that can lead to a striking underestimation of the expected extreme frequency. A simple way of doing this is to remove the mean seasonal cycle before calculating the threshold and subsequent extremes (Brunner and Voigt, 2024). Second, the relative threshold indicators are separated from health-related exposure risks. Since they are based solely on the probabilistic distribution of meteorological data, they fail to account for the physical intensity of extreme weather impacts on human health, resulting in weak correlations between exposure assessments and health outcomes. To address this limitation, some studies have optimised thresholds by integrating health data and thereby improved the health relevance of population exposure assessments. To investigate the additional effects of heatwaves on mortality among Beijing residents from 2007 to 2013, Niu et al. (2022) selected the 90.0th, 92.5th and 95.0th percentiles of daily average temperatures during the study period as thresholds and combined them with durations of 2/3/4 days, resulting in a total of nine heatwave definitions. In addition, Chen et al. (2019) compared 19 definitions of coldwaves when studying coldwaves and mortality in 31 provincial capital cities in China from 2007 to 2013. Third, the relative threshold indicators overlook population vulnerability differences. Under the same percentile threshold, high-vulnerability groups (e.g. older adults, low-income populations) face higher actual exposure risks, which are not captured by the threshold itself (Luo et al., 2025). In the research into social disparities in extreme heat days across US public schools, it was found that racial/ethnic minority students and those eligible for free/reduced lunch face greater risk and disparities observed with both absolute and relative temperature thresholds (Chakraborty and Soroka, 2025). Finally, the relative threshold indicators based on meteorological elements have insufficient ability to distinguish regional health heterogeneity, and this can delay identification of extreme events in high-vulnerability areas while over-identification may occur in low-vulnerability areas (Hosseinzadehtalaei et al., 2025; Chen et al., 2025). Considering regional health heterogeneity, to construct definitions of heatwaves for different regions, Lin et al. (2019) analysed the exposure–response relationship between extreme high temperature and mortality in six different regions of China and obtained unique heatwave temperature thresholds for each region.

In summary, composite threshold indicators can compensate for the above limitations by combining multiple factors and are being applied gradually to various types of research, but the lack of standardisation in these adjustments reduces cross-study comparability. Therefore, the emergence of the composite threshold indicators increases cross-study comparability.

2.3. Composite threshold indicators

Composite threshold indicators are defined as multiple relevant elements or indicators combined to form a new indicator, such as those combining meteorological elements with temporal, spatial and other factors (Li et al., 2013). However, there are certain differences between composite threshold indicators, combined threshold indicators and multi-criteria indices. Composite threshold indicators emphasise 'comprehensiveness' and 'integration', constructing a new indicator through multiple elements, and this new indicator encompasses all the information of multiple elements. Combined threshold indicators merely emphasise consolidation, which does not necessarily construct new indicators. Multi-criteria indices emphasise a multi-dimensional framework, and the information contained in multiple elements used to construct new indicators is in a parallel relationship.

In the research into population exposure assessment, composite threshold indicators offer some critical advantages. First, due to the comprehensive consideration of impact elements, composite threshold indicators can reflect the risk of extreme weather events from more dimensions. For instance, the China Meteorological Administration's grading of heatwaves incorporates temperature and relative humidity to obtain the torridity index (TI), further utilising the duration of the TI to derive the heatwave index (HI), classifying them as mild, moderate and severe. Second, multiple factors can refine the multifaceted nature of the same extreme event. Díaz-Poso et al. (2023) assessed

coldwave intensity in the Iberian Peninsula by deriving two indices: ECIsig for excess coldness quantification by comparing the 3-d average temperature with the 5th percentile of daily mean temperatures in the reference period, and ECIaccl for cold adaptation measurement by comparing ECIsig with the previous 30-d average temperature. These two indices were combined to generate the excess cold factor, classified as low, severe and extreme levels. Third, composite threshold indicators enable population exposure assessment of new or compound extreme events, addressing the gap in single-threshold methods for quantifying exposure in complex scenarios (Liu et al., 2026). For example, Peng et al. (2023) defined four types of compound events based on temperature and precipitation thresholds: warm-wet/dry events and cold-wet/dry events.

Finally, unlike relative thresholds, which can only be associated with health data, the most important advantage is that composite threshold indicators integrating health information allow for the direct quantification of health-relevant exposure risks, supporting targeted public health interventions. The most notable research is the high-temperature health risk early warning tools developed by various countries. The US National Weather Service (NWS) has multiple tools to assess the potential for heat stress due to extreme temperatures, such as the HI, the wet bulb globe temperature (WBGT) and heat risk (NWS, 2025). The HI is a measure of how hot it really feels when relative humidity is factored in with the actual air temperature. The WBGT combines temperature, humidity, wind, solar radiation and other weather parameters and is a particularly effective indicator of heat stress for active populations such as outdoor workers and athletes. The NWS heat risk is based on the high-resolution NWS national gridded forecast database, and a daily heat risk value is calculated for each location from the current date for 7d, with points of concern including: 1) how much above normal the temperatures are at each location; 2) the time of year (for example, whether this is unaccustomed early season heat, or accustomed late season heat); 3) the duration of unusual heat (for example, when temperatures overnight are at levels that would lower heat stress, or if warm overnight temperatures continue to add to heat stress into the next day); and 4) if those temperatures are at levels that pose an elevated risk for heat complications, such as heat stress, based on CDC heat–health thresholds. In China, the national high-temperature health risk early warning system is issued by the National Disease Control and Prevention Administration and China Meteorological Administration (CMA, 2025). It uses the relative temperature threshold method of each region to calculate the risk threshold of each region (considering each region's abnormally high temperature) and calculates the exposure–response relationship between temperature and disease (considering the different applicability of high temperature to the population of each region) to construct a high-temperature health risk early warning model. The average and maximum temperatures on the forecast day are compared and analysed with the thresholds corresponding to each risk level to determine the high-temperature health risk level on the forecast day. Both systems highlight the factors and methods that need to be considered when integrating health information for population exposure assessment to extreme weather events. Consequently, they have an important reference value for future research.

However, the construction of a composite threshold indicator also brings a series of more complex methodological challenges. First, poor policy and public accessibility—due to the complexity of composite indices—reduce the practicality of exposure early warnings. For example, the general public may know HI is level 4 and the heat health risk is high on a given day, but they do not understand the relationship with the meteorological temperature. Additionally, the methodological complexity and high data requirements (e.g. high-resolution meteorological data and health data) limit application in regions, such as low-income countries, with limited data availability.

Overall, the abundant information contained in the composite threshold indicator marks the key transformation of related research from 'weather-centric theory' to 'impact-centric theory'. It changes the focus from 'How abnormal the weather is' to 'How much impact extreme weather will have on population and society', and consequently, it better meets the requirement of population exposure assessment.

3. Methods of population exposure assessment

The IPCC (2012) defines exposure as 'the presence of people, livelihoods, environmental services, various resources, infrastructure, and economic, social, or cultural assets in locations that could be adversely affected'. Population exposure refers specifically to the population residing within the potential impact area of extreme weather events. It is a necessary condition for disaster risk, and calculating and analysing the magnitude of population exposure is a crucial step in assessing the disaster risk associated with extreme weather events. Current assessment methods must balance accuracy and operability. They can be categorised into three types of population exposure: static, dynamic and alternative indicators (Fig. 2).

3.1. Static population exposure

Static population exposure assessment methods are based on the assumption of stable population distribution, quantifying exposure through two core approaches: 1) using the total population size directly within the extreme weather impact area as the exposure indicator (Liu et al., 2023); and 2) calculating the product of extreme weather events' frequency (per grid or administrative unit) and population size to reflect cumulative exposure (Grant et al., 2025; Zhou et al., 2024). The advantages of these

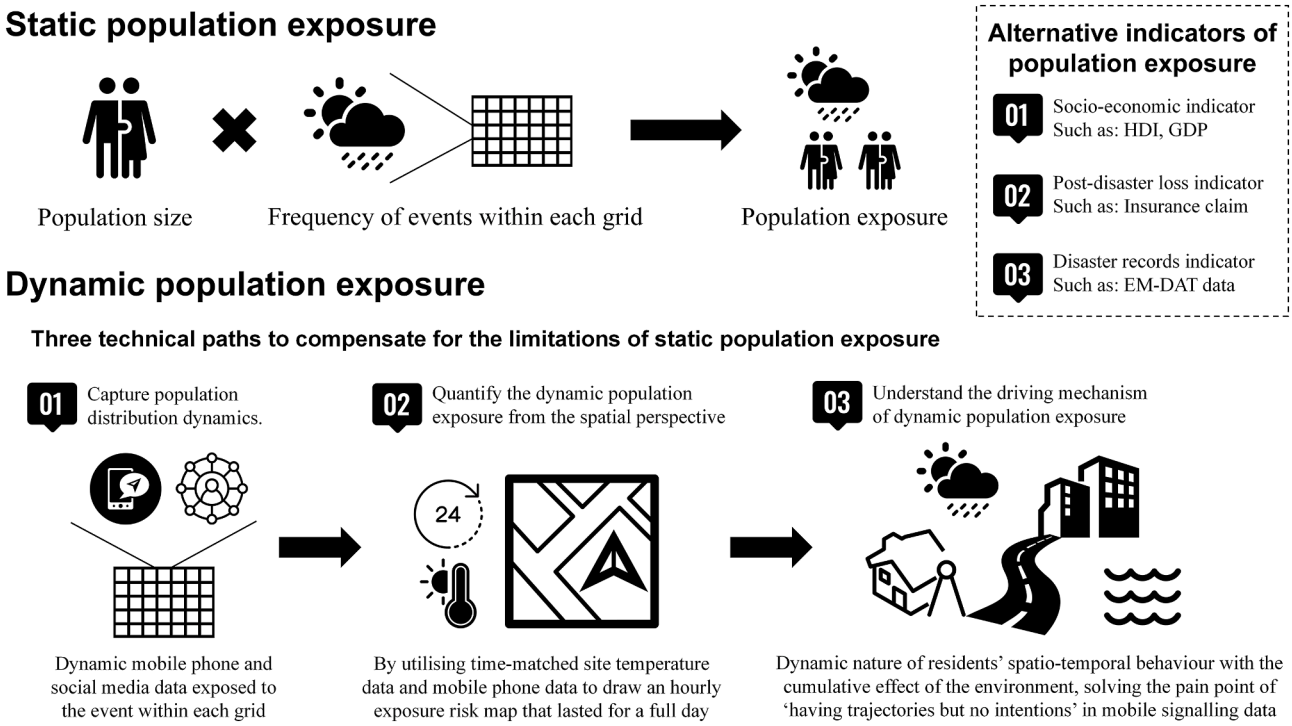


Fig. 2. Population exposure assessment methods.

approaches are that strong operability with low data and technical requirements enable rapid, large-scale exposure screening. Static population exposure is suitable for preliminary risk assessment in regions with limited technical capacity, providing a baseline reference for comparing cross-regional exposure. Meanwhile, the above methods are widely used in macro-level public health planning, such as identifying national or regional high-exposure areas to guide the allocation of disaster prevention and control resources, for instance, using the above methods to predict future increases in extreme precipitation exposure in South China, providing a basis for public health emergency preparedness (Tang and Hu, 2022). However, the static population assumption cannot reflect the duration and intensity of the impact on the region's population exposure, failing to reflect the heterogeneity of exposure risks and lacking consideration of the limitations of population spatiotemporal mobility, resulting in underestimation or overestimation of exposure levels.

3.2. Dynamic population exposure

By integrating spatiotemporal data on population mobility and environmental heterogeneity, dynamic population exposure assessment methods overcome the limitations of static population exposure and represent the core innovation direction in population exposure assessment methods. They are divided into three technical paths.

1) Utilising real-time or near-real-time spatiotemporal data (e.g. mobile phone location data, social media data, public transportation records) to capture population distribution dynamics. For example, Xie et al. (2020) analysed population exposure to heatwaves in Shenzhen based on mobile phone location data. They defined the hourly population in each grid as the sum of population weighting and mobile phone location data. Dynamic population exposure was then defined as the population within grids where temperatures exceeded 35 °C for each hour. While this approach accurately reflects population mobility-induced exposure changes and is suitable for a fine-grained public health dynamic early warning in an urban setting, it faces data privacy risks and technical barriers.
2) Quantifying the dynamic population exposure from the spatial perspective by constructing exposure risk maps. For example, Zhang et al. (2024) utilised time-matched site temperature data and mobile phone signalling data to draw an hourly heat exposure risk map that lasted for a full month. This method confirms the effectiveness and necessity of hourly scale mapping analysis in fully understanding the spatial heterogeneity of urban population heat exposure risks and breaks through the limitation of 'average exposure' in traditional methods, enabling precise identification of high-risk areas (e.g. older communities with limited green space). Furthermore, community-level targeted interventions can be guided. However, it requires complex data fusion and modelling, with high computational costs and limited applicability in regions lacking high-resolution environmental data.

3) Using multi-source data to couple the residents' dynamic visiting behaviour with the cumulative effect of the environment to understand the driving mechanism of dynamic population exposure. For example, Guo et al. (2025) innovatively proposed a novel 'probability–intensity–visitation (PIV) exposure assessment model', which couples the dynamic characteristics of residents' spatiotemporal behaviour (Point of interest [POI] access frequency) to the cumulative effect of the environment (humidity–heat probability × intensity), solving the pain point of 'having trajectories but no intentions' in mobile signalling data and addressing three critical limitations: cumulative impacts, subjective biases and interpretability. Additionally, this method can be used to assess the resilience of human mobility under extreme weather disaster conditions. Tang et al. (2023) focused on the resilient response patterns of population movement behaviours to disasters, and analysed the movement behaviours of populations in the three stages of 'pre-disaster–during disaster–post-disaster' by using various spatiotemporal quantification methods, such as complex network evolution models, to assess exposure risks. However, what is even more difficult than '2)' is that this method requires more complex data fusion.

In summary, the three technical paths respectively addressed the issue of dynamic population exposure: quantification of dynamic population exposure, exposure assessment accuracy and analysis of driving mechanisms.

3.3. Alternative indicator-based population exposure

When direct population or environmental data are unavailable, alternative indicators (indirectly reflecting exposure through correlation with hazard impacts) are used for assessment. First, the socioeconomic indicators, such as the Human Development Index (HDI) and income indices (Russo et al., 2019). Socioeconomic indicators not only indirectly reflect exposure but also embed dimensions of vulnerability (populations in areas with lower levels of development are generally more vulnerable to extreme events), making the assessment more comprehensive. Second, post-disaster loss indicators, such as insurance claim data for disasters caused by local extreme weather events, can serve as an indicator of population exposure risk (Smith and Katz, 2013). A higher per capita claim amount indicates more severe population exposure to extreme weather-induced disasters. Finally, disaster records indicators, such as EM-DAT data (Keim, 2020), can be used to calculate the correlation between the exposure rate dependent variable (e.g. the number of people affected per 100,000) and the hazard occurrence rate independent variable (e.g. the number of extreme events per year) to assess population exposure risk.

These alternative indicators offer a 'suboptimal yet feasible' strategy for population exposure assessment. However, they do not measure the exposed population directly, and the following problems may arise. 1) When groups are intensively exposed to extreme events in high economic areas, the use of socioeconomic indicators may mask such internal differences. 2) High losses do not necessarily represent high exposure, and may also result from extreme vulnerability or improper disaster prevention measures. 3) Disaster data have certain inclusion criteria and usually lose some exposed information. 4) An alternative indicator is usually annual and regional-level, while population exposure is instantaneous and local, leading to a problem of scale mismatch.

4. Future expectations

Current studies have made progress in determining threshold indicators (absolute, relative, composite) and innovating assessment methods (static, dynamic, alternative indicators), but face some challenges. To better assist researchers in using appropriate methods to assess population exposure to extreme weather events, this study offers suggestions in the following three aspects.

4.1. Standardisation and multi-factorial nature of extreme weather definitions

This study identified a groundbreaking point that advanced composite threshold indicators have actually been quietly deconstructing the traditional concept of 'threshold'. It is no longer a single numerical threshold, but a dynamic, multi-factorial risk assessment system. The future development direction of the composite threshold should not blindly pursue more complex indicators, and the following points should be considered: 1) Based on international consensus, we can establish core indicators that integrate meteorological factors, health impacts and regional adaptability for different extreme weather types. Meanwhile, we should avoid over-complexity and prioritise operability to enhance cross-study and cross-regional comparability. 2) The indicators should be tailored to the local context to ensure that they reflect the vulnerability information. It is necessary to develop 'simplified versions' of composite indicators for data-scarce regions to narrow the global assessment gap. 3) Focusing on emerging compound events and constructing multi-factorial composite threshold indicators can be effective in bridging the gap in exposure assessment of non-traditional extreme events.

4.2. Optimisation of population exposure assessment methods

Population exposure serves as an essential reference indicator for evaluating disaster risks posed by extreme events. It is crucial for determining thresholds for certain

factors influencing exposure and constructing comprehensive exposure assessment models. This study found that population exposure will cease to be regarded as a static population existence, but will be understood as a highly dynamic behavioural response process in future studies. However, the following points about the future development direction of the dynamic population exposure should be considered: 1) To address the need of complex data fusion and modelling, future studies can improve models by simplifying parameterisation processes and reducing reliance on multi-source high-resolution data, and verify model applicability across different regions (e.g. cities–rural areas, cross-countries) to promote widespread use. Alternatively, researchers can develop low-cost data collection alternatives (e.g. combining satellite remote sensing with community surveys); 2) For global/regional macro-scale research, optimise static and simplified dynamic methods to ensure efficient large-scale screening; for urban/community micro-scale research, refine high-resolution dynamic assessment models to support fine-grained public health interventions; and 3) Future studies should focus on the cumulative effects of extreme weather on populations and the driving mechanisms of exposure (e.g. spatiotemporal activity intentions, adaptive behaviours). This will achieve the transformation from 'quantifying exposure quantity' to 'explaining exposure causes' to support targeted intervention design.

4.3. Multi-dimensional integration of exposure, vulnerability and adaptability

The above two suggestions are based more on the limitations of methodology, but establishing an 'exposure–vulnerability–adaptability' integrated assessment framework is more important. Notably, distinguishing the connotations of exposure (population distribution in hazard areas) and vulnerability (population sensitivity and coping capacity) is critical to avoid confounding. When embedding vulnerability indicators (age, health status, socioeconomic level) and adaptability indicators (disaster prevention facilities, emergency response capacity) into exposure assessment, studies showed that linking assessment results with health outcomes by integrating health data into composite thresholds and dynamic models can improve public health practice levels effectively.

5. Conclusions

Appropriate assessment of population exposure to extreme weather events is a critical cornerstone of climate-resilient public health systems. This study explored the core questions of the 'why' and 'how' of using appropriate methods to assess population exposure to extreme weather events, finding that existing assessment systems suffer from non-standardised definitions of extreme weather events and inaccurate assessment methods, leading to distorted perceptions of population exposure risk. Thus, by clarifying the necessity of scientific assessment and adopting targeted strategies—promoting the standardisation and multi-factorial transformation of extreme weather event definitions, optimising the methodological system of population exposure assessment and constructing an integrated assessment framework of exposure–vulnerability–adaptability—we can bridge the gap between scientific research and practical application, and improve the scientific rigour and effectiveness of the assessment of population exposure to extreme weather events.

Declaration of competing interest

The authors declare no conflict of interest.

CRediT authorship contribution statement

Ze-Hua Zheng: Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Jing Li:** Writing – review & editing, Writing – original draft, Investigation. **Yuan-Yuan Liu:** Methodology. **Qing Wang:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Tian-Tian Li:** Conceptualization.

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